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Manuscript submitted by co-author Eko Arip Winanto on behalf of all authors.

To: The Editor-in-Chief
IEEE Transactions on Information Forensics and Security

Re: Submission of an original research manuscript

Dear Editor,

We are pleased to submit our manuscript, **"Surviving Sanitization with QuantGhost: Deployment-Robust and Stealthy Steganography in Neural-Network Weights,"** for consideration as an original research article in the *IEEE Transactions on Information Forensics and Security*.

Pre-trained neural networks are increasingly shared through public model hubs, which turns the model file itself into a supply-chain attack surface. Prior weight-steganography attacks (StegoNet, EvilModel, MaleficNet) hide megabyte-scale payloads in float32 weights, but, as we demonstrate on real models, those payloads are completely erased by the routine deployment transforms that production systems apply: INT8 quantization, pruning, and fine-tuning. These are also the very operations the community has proposed as defenses. Our paper asks, and answers, whether a payload can survive the full deployment pipeline while remaining stealthy and accuracy-preserving, and turns the answer into both a fundamental limit and a concrete defense.

The manuscript makes the following contributions:

- **A capacity–robustness–stealth trilemma.** We prove that no weight-embedding scheme can be simultaneously high-capacity, permutation-invariant, and stealthy-quantization-robust, with a tight bound of $\sum_\ell k_h^{(\ell)} k_w^{(\ell)}$ bits on any permutation-invariant channel. This limit governs prior robust watermarking as well as our own attack.
- **QuantGhost.** We present a quantization-aware, distribution-preserving embedding that reaches the high-capacity, stealthy, quantization-robust corner of the trilemma. It survives INT8, pruning, and fine-tuning at 3.2 MB on a real ResNet-50 for only 0.19 percentage points of accuracy, and we validate that it survives byte-exactly through PyTorch's *production* INT8 kernel, not merely an analytical quantizer.
- **An attack and its detector.** Against the released Model X-Ray steganalyzer QuantGhost is statistically indistinguishable from benign weights on a diverse model population, where the prior attack is detected. We then propose a targeted sub-quantum residual detector that reliably flags the basic attack across a narrow and a diverse 3000-model zoo, while honestly showing that an adaptive attacker reaches chance on a realistic population, so detection must be backed by a structural bound.
- **Limits of permutation defenses.** A faithful, full-model neuron permutation on a real ResNet-18, verified function-preserving to machine precision over 30 independent permutations, removes the high-capacity payload, yet a 196-bit permutation-invariant channel survives and end-to-end bootstraps full recovery of an AES-encrypted payload, so permutation-based sanitization is incomplete.

- **Generality and reproducibility.** We show the attack is architecture- and domain-agnostic on a real Vision Transformer and through a cross-dataset study on IoMT/IoT intrusion-detection models (CICIoMT, CICIoT), set out the requirements for a complete defense, and release the full code, data, and a one-to-one table/figure-to-script mapping on Zenodo.

We believe the work is a strong fit for the journal. *IEEE Transactions on Information Forensics and Security* is a leading venue for steganography, steganalysis, information hiding, and machine-learning security; our manuscript contributes a proven information-theoretic limit, a new attack, and a new detector for the protection of machine-learning supply chains, with direct relevance to the IoT/IoMT deployments TIFS readers study.

This research is offensive in nature but carried out to strengthen defenses. All payloads in our experiments are encrypted random bytes or an AES key; no live malware is embedded, executed, or distributed. We will responsibly disclose our findings to the authors of the affected detector and defense (Model X-Ray and NeuPerm) prior to publication, and the paper proposes concrete mitigations.

We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submission and declare no competing interests.

Thank you for considering our work. We look forward to the reviewers' comments.

Sincerely,

Kurniabudi
Corresponding author, on behalf of all authors
(Eko Arip Winanto, Mohd Yazid Idris, and Kurniabudi)
Submitted via ScholarOne Manuscripts by Eko Arip Winanto (winanto@graduate.utm.my).